

SUMBAWANGA, Tanzania, United Republic of

WMO: 638810

Lat: 7.967S

Lon: 31.630E

Elev: 1923

StdP: 80.26

Time Zone: 3.00 (E03)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.8	9.8	-0.6	4.6	23.2	1.3	5.3	21.7	10.1	14.3	8.2	18.1	2.2	130	0.325

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10	10.1	27.2	14.6	26.8	14.7	26.1	14.9	18.7	23.0	18.3	22.4	18.0	22.0	3.8	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.5	15.9	19.8	17.1	15.5	19.5	16.8	15.2	19.3	62.3	23.4	60.7	22.8	59.4	22.4	24.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.0	6.4	5.5		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB	5.6	22.0	1.1	2.1	4.8	23.5	4.2	24.7	3.6	25.9	2.8	27.5
			WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.1	19.1	19.3	19.4	19.2	18.5	17.2	16.9	18.2	19.8	21.1	20.7	19.5
	DBStd	1.58	0.94	0.98	1.03	0.80	0.93	0.97	1.17	1.00	1.08	1.06	1.43	1.03
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	120	3	2	3	2	9	35	46	14	2	0	1	2
	CDD10.0	3311	282	262	292	276	263	217	214	254	293	343	322	293
	CDD18.3	389	27	31	36	28	14	2	2	11	46	84	73	37
	CDH23.3	1740	37	61	64	32	51	82	110	183	337	433	291	58
CDH26.7	60	1	1	0	0	1	0	2	7	14	23	13	0	
Wind	WSAvg	2.2	1.2	1.5	1.6	2.3	2.6	2.6	2.8	2.6	2.8	3.0	2.4	1.5
Precipitation	PrecAvg	1096	219	189	197	94	13	1	1	1	4	30	123	225
	PrecMax	1373	309	253	280	153	86	4	5	5	15	76	219	361
	PrecMin	869	160	115	120	32	0	0	0	0	0	8	46	81
	PrecStd	116	40	36	36	34	16	1	1	1	4	20	51	67
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.2	25.9	25.9	25.1	25.5	26.1	26.8	27.4	28.0	28.2	27.9	25.8
		MCWB	17.1	17.1	18.1	17.2	15.1	14.4	13.6	13.5	13.8	14.9	14.4	16.8
	2%	DB	24.2	24.9	25.0	24.2	24.6	25.1	25.2	26.4	27.0	27.2	26.9	24.9
		MCWB	17.2	17.2	17.5	17.0	15.4	14.1	13.3	13.5	14.3	15.1	15.1	16.5
	5%	DB	23.6	24.1	24.1	23.7	23.9	24.4	24.8	25.2	26.2	26.7	26.0	24.0
		MCWB	17.1	17.2	17.3	16.9	15.6	14.1	13.3	13.6	14.4	15.1	15.2	16.7
	10%	DB	22.8	23.2	23.3	23.0	23.1	23.5	23.7	24.3	25.4	25.9	25.1	23.1
		MCWB	16.9	17.2	17.2	16.7	15.5	14.1	13.1	13.7	14.2	15.1	15.4	16.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.9	19.2	19.3	18.7	18.0	16.6	15.9	16.5	17.3	18.6	18.6	18.9
		MCDB	22.4	23.4	23.6	23.0	22.7	23.5	24.2	24.2	24.6	24.3	22.7	23.0
	2%	WB	18.2	18.5	18.6	18.2	17.2	15.7	14.9	15.5	16.3	17.6	17.7	18.1
		MCDB	21.8	22.7	23.0	22.4	22.4	23.2	23.5	23.8	24.5	23.4	21.8	22.0
	5%	WB	17.8	18.0	18.1	17.8	16.6	15.1	14.3	14.7	15.6	17.0	17.2	17.6
		MCDB	21.3	22.0	22.2	21.8	21.8	22.8	22.8	23.3	24.1	23.4	21.4	21.4
	10%	WB	17.5	17.6	17.7	17.4	16.1	14.5	13.7	14.1	14.8	16.4	16.9	17.2
		MCDB	20.9	21.5	21.7	21.3	21.3	22.2	22.1	22.7	23.5	22.9	21.1	20.9
Mean Daily Temperature Range	5% DB	MDBR	7.7	8.3	8.2	8.3	10.4	13.6	14.3	13.3	12.0	10.1	8.7	7.5
		MCDBR	9.1	9.6	9.3	9.4	11.5	14.7	15.6	14.8	13.1	11.3	9.9	8.7
		MCWBR	3.5	3.8	3.8	3.7	4.5	5.9	6.2	5.6	5.1	4.3	3.5	3.0
	5% WB	MCDBR	8.0	8.5	8.6	8.3	9.9	13.6	14.2	13.3	11.7	10.0	8.4	7.7
		MCWBR	3.5	3.8	4.0	3.8	4.3	6.1	6.2	5.6	5.1	4.3	3.4	3.2
Clear-Sky Solar Irradiance	taub	0.354	0.343	0.331	0.303	0.296	0.291	0.304	0.340	0.371	0.387	0.351	0.338	
	taud	2.507	2.561	2.624	2.766	2.690	2.618	2.524	2.398	2.325	2.297	2.461	2.565	
	Ebn at Noon	984	991	988	989	968	958	951	937	935	937	980	998	
	Edh at Noon	114	108	100	83	85	88	99	117	132	139	118	106	
All-Sky Solar Radiation	RadAvg	4.92	5.41	5.50	5.72	5.83	5.76	5.92	6.29	6.62	6.57	5.82	5.16	
	RadStd	0.38	0.29	0.27	0.26	0.23	0.18	0.13	0.16	0.20	0.34	0.36	0.35	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page